

QTT2

Finite Mathematics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A principal surveyed 60 teachers. The survey was emailed on a Monday and closed after 10 days. 35 teachers coach a sport, 28 advise a club, and 12 do both. How many teachers do neither?

- A. -3, so the data must be wrong
- B. 9
- C. 12
- D. 51

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Sets A and B have $|A| = 32$, $|B| = 30$, and $|A \cup B| = 50$. How many elements are in $A \cap B$?

- A. 62
- B. 12
- C. 18
- D. It cannot be found without a Venn diagram.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A depot with 14 buses runs one route every 12 minutes and another every 18 minutes. Both leave at 7:00 a.m. The depot was repaved last year. When will both routes next leave at the same time?

- A. 7:30 a.m.
- B. 10:36 a.m.
- C. 7:06 a.m.
- D. 7:36 a.m.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A teacher has 48 pencils and 72 erasers. She wants identical kits that use every item. What is the greatest number of kits she can make?

- A.** 120
- B.** 12
- C.** 24
- D.** 144

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A student in a class of 26 wrote $18 - 6 \div 3 \times 2 + 4^2$ on the board during a review day. What is its value?

- A.** 33
- B.** 24
- C.** 30
- D.** 22

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Today is Tuesday. A school that opened 12 days ago plans a festival exactly 100 days from today. On what day of the week will the festival fall?

- A.** Saturday
- B.** Tuesday
- C.** Thursday
- D.** Sunday

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An auditorium that was repainted in June has 14 seats in the first row, and each row has 3 more seats than the row in front. How many seats are in row 20?

- A.** 74
- B.** 71
- C.** 14×3^{19}
- D.** 850

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

In the geometric sequence 5, 15, 45, ..., which term is 3,645?

- A.** The 7th term
- B.** The 365th term
- C.** The 6th term
- D.** The 729th term

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A coach with 11 years on the job posts: "If it rains, then the game is canceled." Parents in the team chat ask when her statement would turn out to be false. Which case makes it false?

- A.** Any day it does not rain.
- B.** It does not rain and the game is not canceled.
- C.** It does not rain and the game is canceled.
- D.** It rains and the game is not canceled.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which statement is logically equivalent to "If a number is divisible by 6, then it is even"?

- A.** If a number is not even, then it is not divisible by 6.
- B.** If a number is not divisible by 6, then it is not even.
- C.** If a number is even, then it is divisible by 6.
- D.** A number is divisible by 6 and even.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A mail carrier with a 3-hour shift limit wants to walk every street exactly once. The street map has six intersections with degrees 2, 4, 3, 3, 2, and 4. What is true?

- A.** An Euler path exists, starting at one degree-3 intersection and ending at the other, but no Euler circuit.
- B.** Neither exists, because some intersections have odd degree.
- C.** An Euler circuit exists because the degrees sum to an even number.
- D.** An Euler circuit exists if the carrier starts at a degree-4 intersection.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An IT team must connect 9 campus buildings with cable so every building can reach every other, using the fewest links and no loops. Each cable run costs between \$400 and \$1,100. How many links are needed?

- A. 9
- B. 72
- C. 36
- D. 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A principal surveyed 60 teachers. ~~The survey was emailed on a Monday and closed after 10 days.~~ 35 teachers coach a sport, 28 advise a club, and 12 do both. How many teachers do neither?

Target word: count of neither **Decides:** $35 + 28 - 12 = 51$ do at least one.

A	Dropped element	$60 - 63$ drops the overlap subtraction.
B	Correct	$60 - 51 = 9$.
C	Wrong stem number	That is the number who do both.
D	Confusion set	That is the number who do at least one.

2. Answer B Concept 1 • transfer

Sets A and B have $|A| = 32$, $|B| = 30$, and $|A \cup B| = 50$. How many elements are in $A \cap B$?

Target word: solve for the overlap **Decides:** $50 = 32 + 30 - |A \cap B|$.

A	Dropped element	Just adds $ A $ and $ B $.
B	Correct	$62 - 50 = 12$.
C	Adjacent category	$50 - 32 = 18$ is the count in B only.
D	Absolute language	The formula alone gives the overlap.

3. Answer D Concept 2

A depot ~~with 14 buses~~ runs one route every 12 minutes and another every 18 minutes. Both leave at 7:00 a.m. ~~The depot was repaved last year.~~ When will both routes next leave at the same time?

Target word: next common departure **Decides:** $\text{LCM}(12, 18) = 36$ minutes.

A	Inverted operation	Adds $12 + 18$ instead of finding a common multiple.
B	Dropped element	$12 \times 18 = 216$ minutes is a common multiple, but it drops the shared factor, so it is not the least.
C	Confusion set	Uses the GCD, 6.
D	Correct	36 is the least common multiple of 12 and 18.

4. Answer C Concept 2 • transfer

A teacher has 48 pencils and 72 erasers. She wants identical kits that use every item. What is the greatest number of kits she can make?

Target word: largest equal grouping **Decides:** $\text{GCD}(48, 72) = 24$.

A	Inverted operation	Adds the counts.
B	Dropped element	A common factor, but not the greatest.
C	Correct	24 kits of 2 pencils and 3 erasers.
D	Confusion set	That is the LCM.

5. Answer C Concept 3

A student in a class of 26 wrote $18 - 6 \div 3 \times 2 + 4^2$ on the board during a review day. What is its value?

Target word: evaluate **Decides:** $4^2 = 16$; then $6 \div 3 \times 2 = 4$ left to right; $18 - 4 + 16$.

A	Inverted operation	Multiplies 3×2 first, then $6 \div 6 = 1$: $18 - 1 + 16$.
B	Dropped element	Works left to right and drops division's precedence over subtraction: $(18 - 6) \div 3 \times 2 + 16$.
C	Correct	$18 - 4 + 16 = 30$.
D	Confusion set	Reads 4^2 as $4 \times 2 = 8$.

6. Answer C Concept 4

Today is Tuesday. A school that opened 12 days ago plans a festival exactly 100 days from today. On what day of the week will the festival fall?

Target word: day of week **Decides:** $100 = 7 \times 14 + 2$, so $100 \equiv 2 \pmod{7}$.

A	Camouflage	Subtracts the school's 12 days: $88 \equiv 4 \pmod{7}$; the opening date does not matter.
B	Confusion set	Treats 100 as a multiple of 7.
C	Correct	Two days after Tuesday.
D	Inverted operation	Counts 2 days backward.

7. Answer B Concept 5

An auditorium that was repainted in June has 14 seats in the first row, and each row has 3 more seats than the row in front. How many seats are in row 20?

Target word: 20th term **Decides:** $a_{20} = 14 + 19 \times 3$.

A	Dropped element	Uses 20×3 and drops the $n - 1$.
B	Correct	$14 + 57 = 71$.
C	Confusion set	Treats the sequence as geometric.
D	Adjacent category	That is the total seats in 20 rows, not row 20.

8. Answer A Concept 5 · transfer

In the geometric sequence 5, 15, 45, ..., which term is 3,645?

Target word: solve for n **Decides:** $5 \cdot 3^{(n-1)} = 3,645$, so $3^{(n-1)} = 729 = 3^6$.

A	Correct	$n - 1 = 6$, so $n = 7$.
B	Adjacent category	Treats it as arithmetic with difference 10.
C	Dropped element	Reports $n - 1$ as n.
D	Confusion set	729 is $3^{(n-1)}$, not n.

9. Answer D Concept 6

A coach ~~with 11 years on the job~~ posts: "If it rains, then the game is canceled." ~~Parents in the team chat ask~~ when her statement would turn out to be false. Which case makes it false?

Target word: false case of $p \rightarrow q$ **Decides:** False only when p is true and q is false.

A	Absolute language	No case with p false can make the conditional false.
B	Confusion set	Both parts false still makes $p \rightarrow q$ true.
C	Inverted operation	A false p makes the conditional true.
D	Correct	The promise is broken only when p holds and q fails.

10. Answer A Concept 6 • transfer

Which statement is logically equivalent to "If a number is divisible by 6, then it is even"?

Target word: equivalent form **Decides:** The contrapositive is equivalent.

A	Correct	The contrapositive always has the same truth value.
B	Confusion set	The inverse; 4 again breaks it.
C	Inverted operation	The converse; 4 is even but not divisible by 6.
D	Adjacent category	A conjunction, not a conditional.

11. Answer A Concept 7

A mail carrier ~~with a 3-hour shift limit~~ wants to walk every street exactly once. The street map has six intersections with degrees 2, 4, 3, 3, 2, and 4. What is true?

Target word: Euler path vs. circuit **Decides:** Exactly two vertices have odd degree.

A	Correct	Two odd vertices allow a path that must start and end at them.
B	Absolute language	Exactly two odd vertices still allow a path.
C	Confusion set	Every graph's degrees sum to an even number; circuits need every degree even.
D	Adjacent category	The start point cannot fix odd degrees; circuits need all even.

12. Answer D Concept 8

An IT team must connect 9 campus buildings with cable so every building can reach every other, using the fewest links and no loops. ~~Each cable run costs between \$400 and \$1,100.~~ How many links are needed?

Target word: edges in a spanning tree **Decides:** A tree on n vertices has $n - 1$ edges.

A	Wrong stem number	One link per building would create a loop.
B	Confusion set	9×8 counts ordered pairs.
C	Adjacent category	That is every possible pair, a complete graph.
D	Correct	$9 - 1 = 8$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Inclusion-exclusion for two sets	$ A \cup B = A + B - A \cap B $; “neither” = total – $ A \cup B $.	Adding the two sets without subtracting the overlap.	Two groups with some members in both.
2	GCD vs. LCM in word problems	“When do they line up again?” is the LCM; “largest equal groups with nothing left over” is the GCD.	Using the GCD for a repeating-schedule question, or multiplying the numbers.	Repeating cycles (LCM) or splitting into identical kits (GCD).
3	Order of operations	Parentheses, exponents, then multiplication and division left to right, then addition and subtraction left to right.	Doing all multiplication before division, or working strictly left to right.	A string mixing $\div$, $\times$, $-$, and a power.
4	Modular (clock) arithmetic	To find a day of the week n days ahead, use $n \bmod 7$ and count forward that many days.	Counting backward, or treating a round number as a multiple of 7.	“What day of the week will it be in ... days?”
5	Arithmetic and geometric sequences	Arithmetic: $a_n = a_1 + (n - 1)d$. Geometric: $a_n = a_1 \cdot r^{(n - 1)}$.	Using n instead of $n - 1$, or mixing the two formulas.	A first term and a constant difference or ratio.
6	Conditional statements	$p \rightarrow q$ is false only when p is true and q is false; it is equivalent to its contrapositive, not its converse or inverse.	Treating the converse as equivalent.	An “if ... then ...” claim and a request for a false case or an equivalent statement.
7	Euler paths and circuits	A connected graph has an Euler circuit if every vertex has even degree, and an Euler path (not circuit) if exactly two vertices have odd degree.	Concluding “neither” as soon as any vertex is odd.	A route that must use every edge (street) exactly once.
8	Trees and spanning trees	A tree on n vertices has exactly $n - 1$ edges; a spanning tree connects all n vertices with $n - 1$ edges.	Using n edges, or counting all possible pairs.	Connect every site with the fewest links and no loops.

Course scope (public description): Finite Mathematics covers the knowledge and skills necessary to apply discrete mathematics and properties of number systems to model and solve real-life problems. Topics include sets and operations; prime and composite numbers; GCD and LCM; order of operations; ordering numbers; mathematical systems including modular arithmetic, arithmetic and geometric sequences, ratio and proportion, subsets of real numbers, logic and truth tables, graphs, and trees and networks. There are no prerequisites for this course.